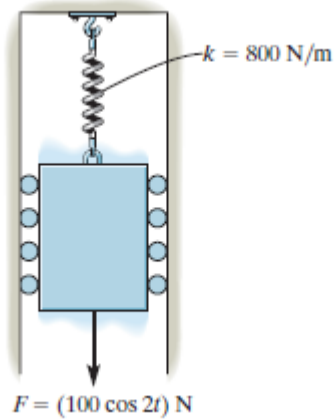


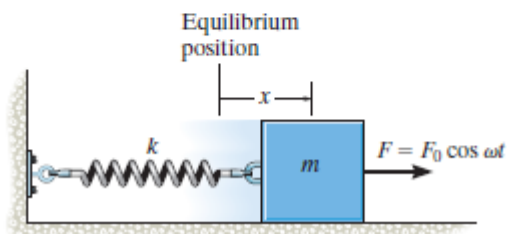
SPRING 2025-2026 – MECT2222 - EXERCISE QUESTIONS FOR THE FINAL EXAM

(ALL SOLUTIONS MUST INCLUDE A FREE-BODY DIAGRAM.)

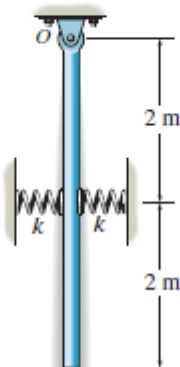
*22–56. The 40-kg block is attached to a spring having a stiffness of 800 N/m. A force $F = (100 \cos 2t)$ N, where t is in seconds is applied to the block. Determine the maximum speed of the block for the steady-state vibration.



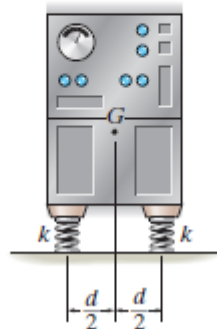
22–42. If the block-and-spring model is subjected to the periodic force $F = F_0 \cos \omega t$, show that the differential equation of motion is $\ddot{x} + (k/m)x = (F_0/m) \cos \omega t$, where x is measured from the equilibrium position of the block. What is the general solution of this equation?



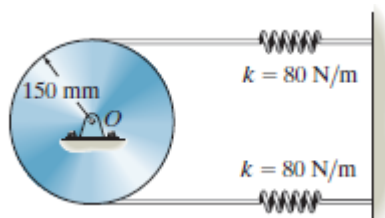
22–37. If the lower end of the 6-kg slender rod is displaced a small amount and released from rest, determine the natural frequency of vibration. Each spring has a stiffness of $k = 200$ N/m and is unstretched when the rod is hanging vertically.



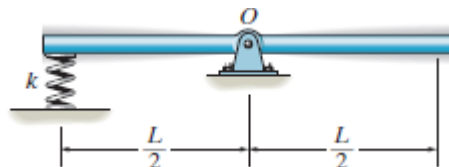
***22–36.** The machine has a mass m and is uniformly supported by *four* springs, each having a stiffness k . Determine the natural period of vertical vibration.



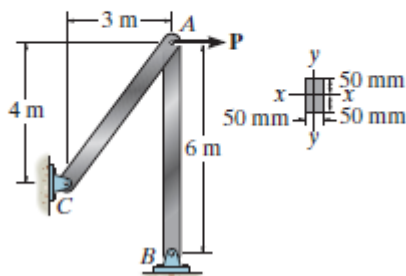
22–23. The 10-kg disk is pin connected at its mass center. Determine the natural period of vibration of the disk if the springs have sufficient tension in them to prevent the cord from slipping on the disk as it oscillates. *Hint:* Assume that the initial stretch in each spring is δ_0 .



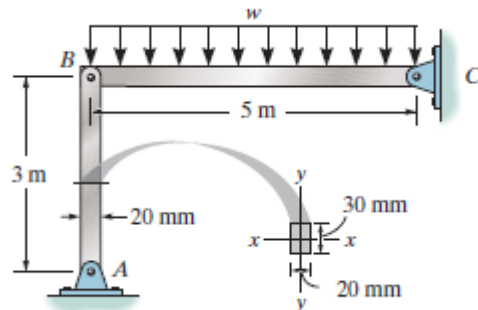
22–15. Determine the natural period of vibration of the uniform bar of mass m when it is displaced downward slightly and released.



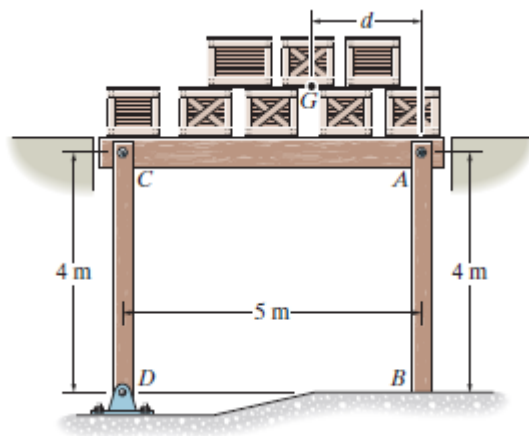
***13–40.** The steel bar AB of the frame is assumed to be pin connected at its ends for y - y axis buckling. If $P = 18 \text{ kN}$, determine the factor of safety with respect to buckling about the y - y axis. $E_{st} = 200 \text{ GPa}$, $\sigma_Y = 360 \text{ MPa}$.



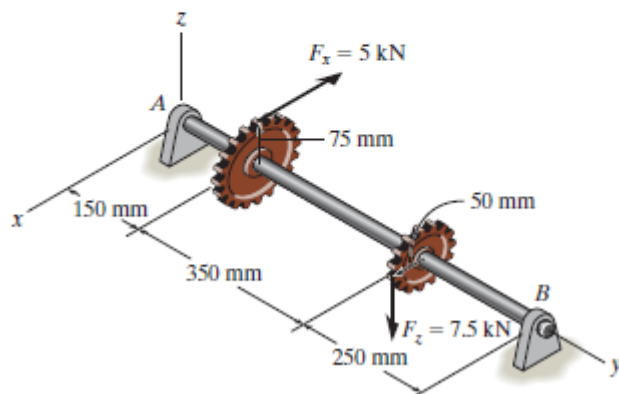
13-34. The steel bar AB has a rectangular cross section. If it is pin connected at its ends, determine the maximum allowable intensity w of the distributed load that can be applied to BC without causing AB to buckle. Use a factor of safety with respect to buckling of 1.5. $E_{st} = 200 \text{ GPa}$, $\sigma_Y = 360 \text{ MPa}$.



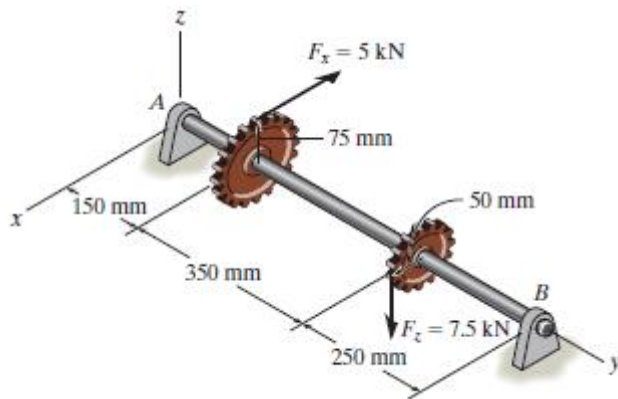
13-13. The deck is supported by the two 40-mm-square columns. Column AB is pinned at A and fixed at B , whereas CD is pinned at C and D . If the deck is prevented from sidesway, determine the position d of the center of gravity of the load and the load's greatest magnitude without causing the deck to collapse. Both columns are made from Douglas Fir.



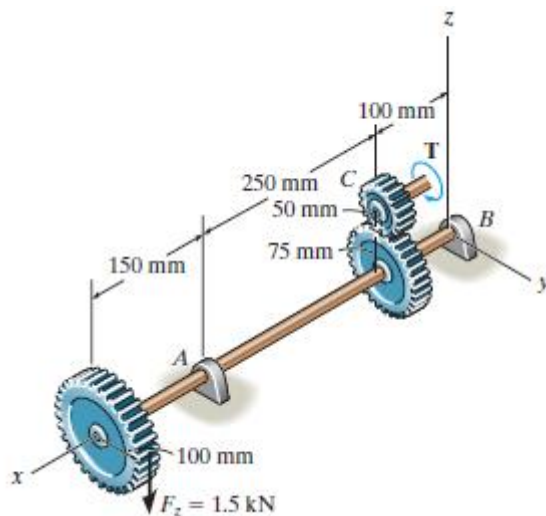
***R11-4.** The journal bearings at A and B exert only x and z components of force on the shaft. Determine the shaft's diameter to the nearest millimeter so that it can resist the loadings. Use the maximum distortion energy theory of failure with $\sigma_{allow} = 200 \text{ MPa}$.



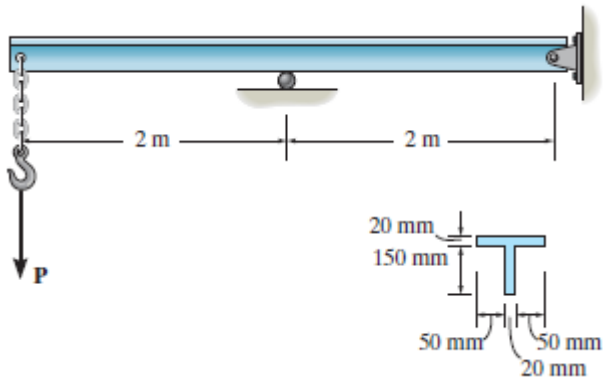
R11-3. The journal bearings at A and B exert only x and z components of force on the shaft. Determine the shaft's diameter to the nearest millimeter so that it can resist the loadings without exceeding an allowable shear stress of $\tau_{\text{allow}} = 80 \text{ MPa}$. Use the maximum shear stress theory of failure.



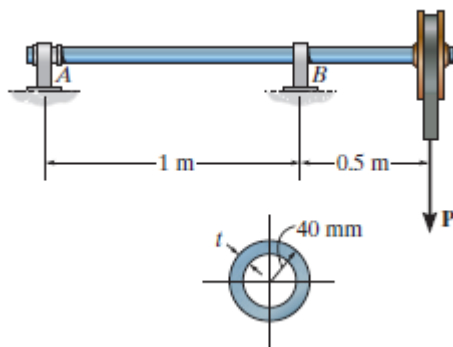
11-43. The end gear connected to the shaft is subjected to the loading shown. If the bearings at A and B exert only y and z components of force on the shaft, determine the equilibrium torque T at gear C and then determine the smallest diameter of the shaft to the nearest millimeter that will support the loading. Use the maximum-distortion-energy theory of failure with $\sigma_{\text{allow}} = 80 \text{ MPa}$.



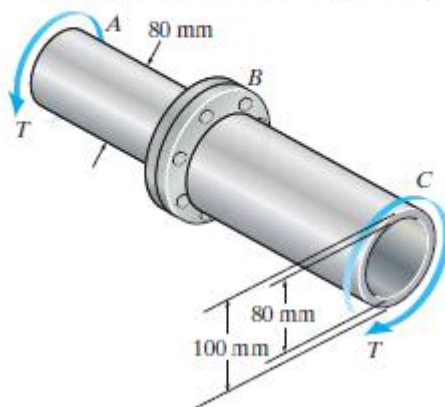
11-30. The steel beam has an allowable bending stress $\sigma_{\text{allow}} = 140 \text{ MPa}$ and an allowable shear stress of $\tau_{\text{allow}} = 90 \text{ MPa}$. Determine the maximum load that can safely be supported.



***11-20.** The shaft is supported by a smooth thrust bearing at A and a smooth journal bearing at B . If $P = 5 \text{ kN}$ and the shaft is made from steel having an allowable normal stress of $\sigma_{\text{allow}} = 150 \text{ MPa}$ and an allowable shear stress of $\tau_{\text{allow}} = 85 \text{ MPa}$, determine the required minimum wall thickness t of the shaft to the nearest millimeter to safely support the load.



10-87. The shaft consists of a solid segment AB and a hollow segment BC , which are rigidly joined by the coupling at B . If the shaft is made from A-36 steel, determine the maximum torque T that can be applied according to the maximum distortion energy theory. Use a factor of safety of 1.5 against yielding.

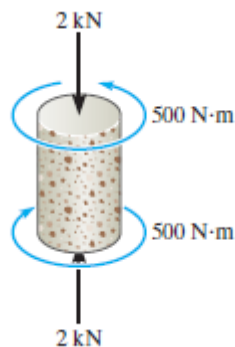


10-73. The state of stress acting at a critical point on the seat frame of an automobile during a crash is shown in the figure. Determine the smallest yield stress for a steel that can be selected for the member, based on the maximum-shear-stress theory.

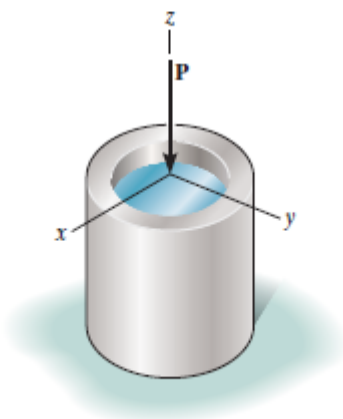
10-74. Solve Prob. 10-73 using the maximum-distortion-energy theory.



10-69. The short concrete cylinder having a diameter of 50 mm is subjected to a torque of 500 N·m and an axial compressive force of 2 kN. Determine if it fails according to the maximum normal stress theory. The ultimate stress of the concrete is $\sigma_{ult} = 28$ MPa.

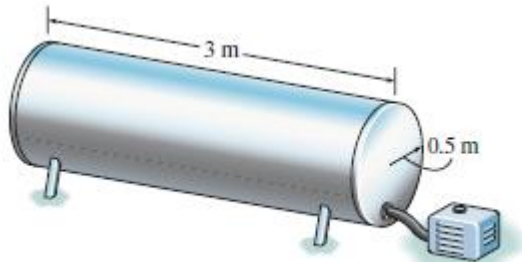


10-58. A soft material is placed within the confines of a rigid cylinder which rests on a rigid support. Assuming that $\epsilon_x = 0$ and $\epsilon_y = 0$, determine the factor by which the stiffness of the material, or the apparent modulus of elasticity, will be increased when a load is applied, if $\nu = 0.3$ for the material.

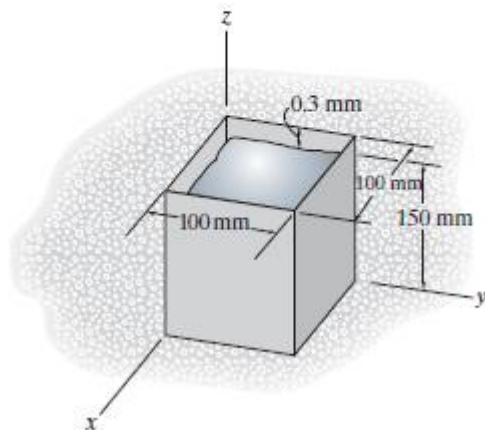


***10–56.** The thin-walled cylindrical pressure vessel of inner radius r and thickness t is subjected to an internal pressure p . If the material constants are E and ν , determine the strains in the circumferential and longitudinal directions. Using these results, calculate the increase in both the diameter and the length of a steel pressure vessel filled with air and having an internal gage pressure of 15 MPa. The vessel is 3 m long, and has an inner radius of 0.5 m and a thickness of 10 mm. $E_{st} = 200 \text{ GPa}$, $\nu_{st} = 0.3$.

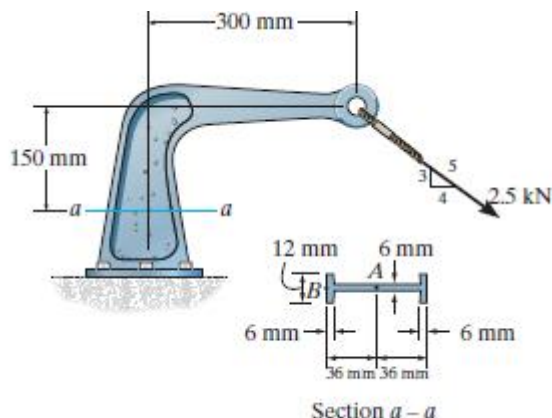
10–57. Estimate the increase in volume of the pressure vessel in Prob. 10–56.



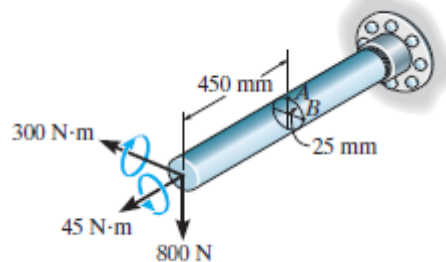
10–41. The smooth rigid-body cavity is filled with liquid 6061-T6 aluminum. When cooled it is 0.3 mm from the top of the cavity. If the top of the cavity is not covered and the temperature is increased by 110°C , determine the strain components ϵ_x , ϵ_y , and ϵ_z in the aluminum. *Hint:* Use Eqs. 10–18 with an additional strain term of $\alpha\Delta T$ (Eq. 4–4).



***9–88.** Determine the principal stress and absolute maximum shear stress developed at point A on the cross section of the bracket at section $a-a$.

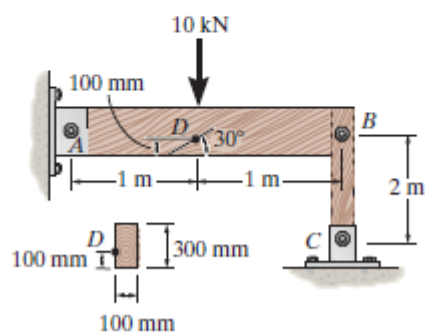


9-85. The solid shaft is subjected to a torque, bending moment, and shear force. Determine the principal stresses at points *A* and *B* and the absolute maximum shear stress.

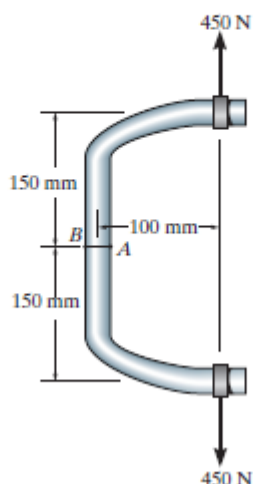


9-74. Determine the normal and shear stresses at point *D* that act perpendicular and parallel, respectively, to the grains. The grains at this point make an angle of 30° with the horizontal as shown. Point *D* is located just to the left of the 10-kN force.

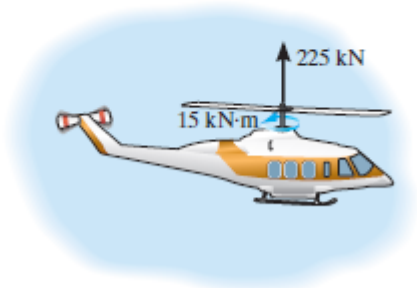
9-75. Determine the principal stress at point *D*, which is located just to the left of the 10-kN force.



9-66. Determine the principal stresses and the maximum in-plane shear stress that are developed at point *A*. Show the results on an element located at this point. The rod has a diameter of 40 mm.

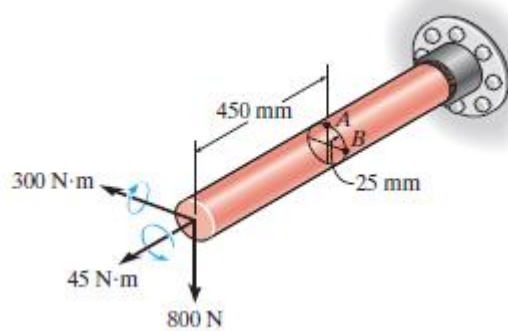


9-63. The rotor shaft of the helicopter is subjected to the tensile force and torque shown when the rotor blades provide the lifting force to suspend the helicopter at midair. If the shaft has a diameter of 150 mm, determine the principal stress and maximum in-plane shear stress at a point located on the surface of the shaft.

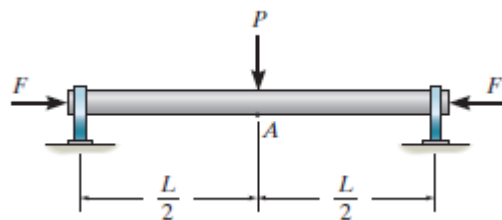


9-38. The solid shaft is subjected to a torque, bending moment, and shear force as shown. Determine the principal stresses acting at point *A*.

9-39. Solve Prob. 9-38 for point *B*.

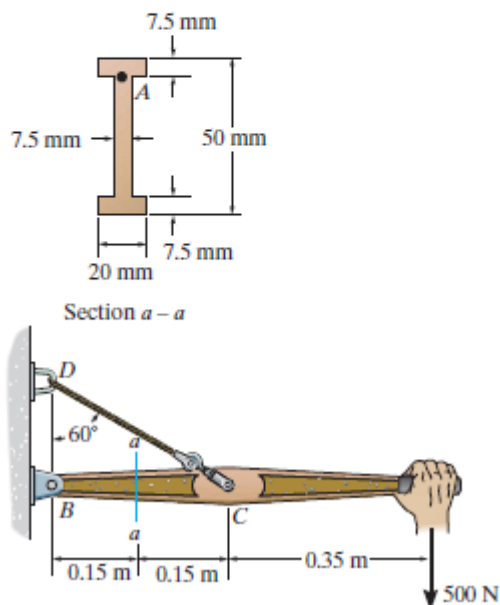


9-37. The shaft has a diameter d and is subjected to the loadings shown. Determine the principal stresses and the maximum in-plane shear stress at point *A*. The bearings only support vertical reactions.

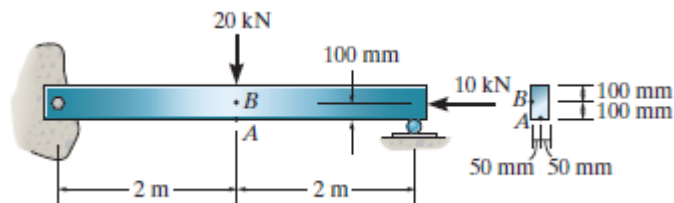


9-34. Determine the principal stress at point *A* on the cross section of the arm at section *a-a*. Specify the orientation of this state of stress and indicate the results on an element at the point.

9-35. Determine the maximum in-plane shear stress developed at point *A* on the cross section of the arm at section *a-a*. Specify the orientation of this state of stress and indicate the results on an element at the point.

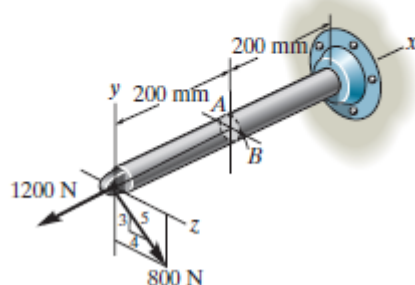


9-30. The beam has a rectangular cross section and is subjected to the loadings shown. Determine the principal stresses at point *A* and point *B*, which are located just to the left of the 20-kN load. Show the results on elements located at these points.

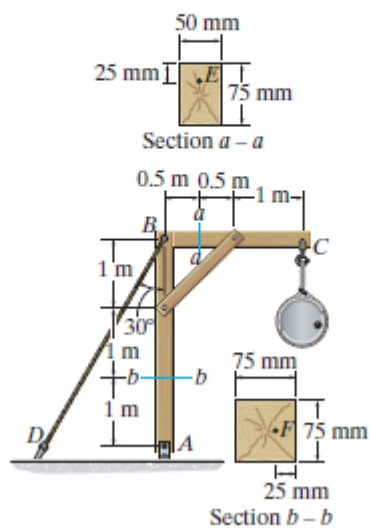


***8-68.** The bar has a diameter of 40 mm. Determine the state of stress at point *A* and show the results on a differential volume element located at this point.

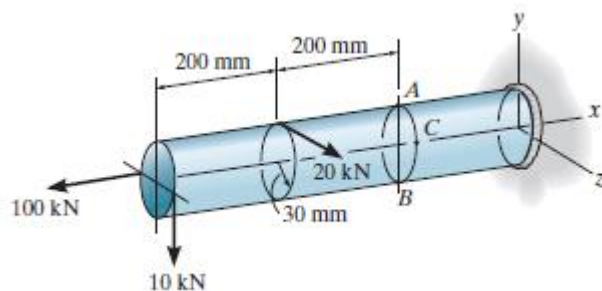
8-69. Solve Prob. 8-68 for point *B*.



R8-3. The 20-kg drum is suspended from the hook mounted on the wooden frame. Determine the state of stress at point F on the cross section of the frame at section $b-b$. Indicate the results on an element.

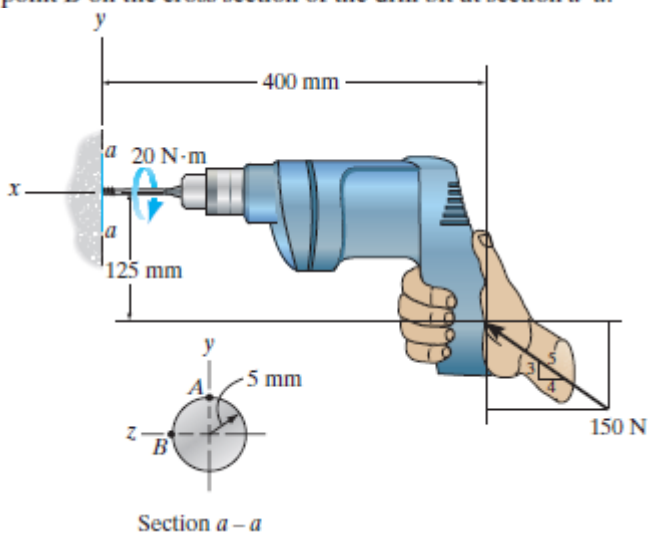


8-49. The solid rod is subjected to the loading shown. Determine the state of stress at point C , and show the results on a differential volume element at this point.

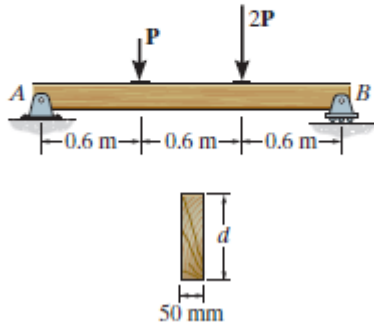


8-31. The drill is jammed in the wall and is subjected to the torque and force shown. Determine the state of stress at point A on the cross section of the drill bit at section $a-a$.

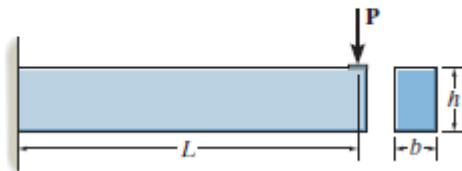
***8-32.** The drill is jammed in the wall and is subjected to the torque and force shown. Determine the state of stress at point B on the cross section of the drill bit at section $a-a$.



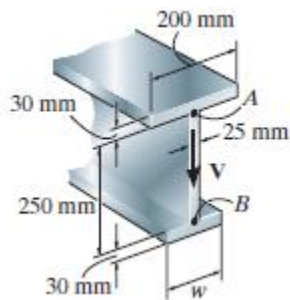
7-26. If the beam is made from wood having an allowable shear stress $\tau_{allow} = 3 \text{ MPa}$, determine the maximum magnitude of P . Set $d = 100 \text{ mm}$.



7-25. Determine the length of the cantilevered beam so that the maximum bending stress in the beam is equivalent to the maximum shear stress.



7-22. If the beam is subjected to a shear of $V = 15 \text{ kN}$, determine the web's shear stress at A and B . Indicate the shear-stress components on a volume element located at these points. Set $w = 125 \text{ mm}$. Show that the neutral axis is located at $\bar{y} = 0.1747 \text{ m}$ from the bottom and $I_{NA} = 0.2182(10^{-3}) \text{ m}^4$.



7-18. Plot the shear-stress distribution over the cross section of a rod that has a radius c . By what factor is the maximum shear stress greater than the average shear stress acting over the cross section?

